jQuery Tutorial

What are we Talking about today?

- Introduction to jQuery
  - Purpose: JS library for simplifying client-side scripting.
  - Core Features: DOM manipulation, event handling, etc.
  - Cross-browser Compatibility
- Key Functionalities
  - Selectors: Simplifies selection/ manipulation of DOM elements
  - Event Handling: Enables responsive interactions with methods
  - AJAX Support
- Benefits and Considerations
  - Write Less, Do More
  - Plugin Support
  - Modern Alternatives: Newer frameworks and native JavaScript

What is jQuery?

- A framework for client-side JavaScript.
- Frameworks provide useful alternatives for common programming tasks.
- An open-source project at jquery.com
- It simplifies
  - HTML document traversing
  - Event Handling
  - Animating
  - AJAX interactions

What is available with jQuery?

- Cross browser support and detection
- AJAX functions
- CSS functions
- DOM manipulation
- DOM transversal
- Attribute manipulation
- Event detection and handling
- JavaScript animation
- Hundreds of plugins for pre-built user interfaces, advanced animations, form validation, etc
- Expandable functionality using custom plugins
- Small foot print

Downloading jQuery

- Installation – You just download the latest jquery-x.y.z.js file and put it in your website folder
- http://jquery.com/download
- jQuery is lightweight:
  - 90KB (minified and uncompressed, jquery-3.7.1.min.js)
  - 33KB (Minified and Gzipped)
- Latest version is 3.7.1 (4.0 in beta)

So How Does jQuery Change How You Write JavaScript?

- jQuery adds a JavaScript object called **$** or **jQuery** to your JavaScript code.
  - Through manipulation of this JavaScript code, it abstracts away commonly used JavaScript objects into $ and jQuery, such as the DOM (document), XMLHTTPRequest, and JSON
- Example: Instead of

```
var myButton = document.getElementById("myButton");
```

- In jQuery, it’s just

```
$("#myButton");
```

Modern DOM API

- However, all modern browsers supports similar functionalities:
- Querying:

JQuery: `$("…")`
DOM API: `node.querySelector` / `node.querySelectorAll`

This supports all valid css selectors:

```
"#id", ".class", "node", "#nested > .btn:first-child"
```

The following replacement could be used to simplify code

```
const $ = document.querySelectorAll
```

- See:

https://developer.mozilla.org/en-US/docs/Web/API/Element/querySelector
https://developer.mozilla.org/en-US/docs/Web/API/Element/querySelectorAll

jQuery Category: Basic Selectors

- These are examples of "Basic" selectors, based on CSS1:
- **All Selector ("*"):** selects all elements, sets css properties and returns the number of elements found

```
var elementCount = $("*").css("border", "3px solid red" ).length;
```

- **Class Selector (".class"):** selects all elements with a given class and sets css properties

```
$(".myClass").css("border", "3px solid red");
```

- **Element selector ("element"):** selects all elements with the given tag name, e.g. div, and sets css properties

```
$("div").css("border", "9px solid red");
```

- **ID selector ("#id"):** selects a single element with the given id attribute

```
$("#myDiv").css("border", "3px solid red");
```

- **Multiple selector ("selector1, selector2, selectorN"):** selects a combined result of all the specified selectors

```
$("div, span, p.myClass").css("border", "3px solid red");
```

- For more examples see: http://api.jquery.com/category/selectors/basic-css-selectors/

Other jQuery Selector Categories

- JQuery borrows notation from CSS1-3 "selectors", as a tool to match a set of elements. Here are some examples of what one can do:
- Attribute: selects elements that have the specified attribute and changes the associated text

```
$("input[value='Hot Fuzz'] ").text("Hot Fuzz");
```

- Basic Filter, e.g. selects all elements that are h1, h2, h3, etc and assigns css properties

```
$(":header").css({ background: "#ccc", color: "blue" });
```

- Child Filter, e.g. finds the first span in each div and underlines the text

```
$("div span:first-child").css("text-decoration", "underline");
```

- Content Filter, e.g. finds all div containing "John" and underlines them

```
$("div:contains('John')").css("text-decoration", "underline");
```

- Form, e.g. finds all buttons and adds the css class "marked" to their properties

```
var input = $(":button").addClass("marked");
```

- For more examples see: http://api.jquery.com/category/selectors

jQuery Functions (see Events in API docs)

- Either attached to the jQuery object or chained off of a selector statement (see *Category: Events – Document Loading*).
  - E.g., Run a function when the page is fully loaded

```
$(window).on("load", function () {
    //run code
});
```

- Most functions return the jQuery object they were originally passed, so you can perform many actions in a single line.
  - E.g., Add the class *bigImg* to all images with height > 100 once the image is loaded (see *Category:Attributes*)

```
$("img.userIcon").load(function () {
    if ($(this).height() > 100) {
        $(this).addClass("bigImg");
    }
});
```

- The same function can perform an entirely different action based on the number and type of parameters.

More jQuery Examples

- Remember these examples?

  http://csci571.com/examples.html#dom

DOM Examples

- Example 1 ➔ document.getElementById.style.color
- Example 2 ➔ document.getElementsByTagName
- Example 3 ➔ document.getElementById().innerHTML
- Example 4 ➔ Moving Objects Horizontally
- Example 5 ➔ Reversing Nodes in a Document
- Example 6 ➔ DOM and Three innerhtml Examples
- Example 7 ➔ DOM setting CSS Background Property
- Example 8 ➔ DOM setting CSS Background Image Property
- Example 9 ➔ DOM used for switching stylesheets

- We’ll revisit the examples, but with jQuery instead!

  https://csci571.com/examples.html#jquery

Example 1: document.getElementById.style.color

John slowly faded into view.

Fade Text

JavaScript w/o jQuery

```
let hex = 255; // Initial color value.

function fadetext() {
  if (hex < 0) return (hex = 255); //reset hex value

  hex -= 11; // increase color darkness
  const el = document.getElementById("sample");
  el.style.color = `rgb(${hex},${hex},${hex})`;
  setTimeout(() => fadetext(), 20);
}
```

http://csci571.com/examples/dom/ex1.html

Example 1: $.fadeOut(), $.delay(), $.fadeIn()

John slowly faded into view.

Fade Text

JavaScript with jQuery

```
// when document is ready
$(function () {
  // set a onClick handler on fadeText
  $("#fadeText").click(function () {
    // fadeOut the h3 for 125 ms, delay, then fadeIn
    $("h3").fadeOut(125).delay().fadeIn(125); \
  });
});
```

http://csci571.com/examples/jquery/dom/ex1.html

Example 2: document.querySelectorAll

Font1
Font2
Font3
Font4
Count Font Tags

JavaScript w/o jQuery

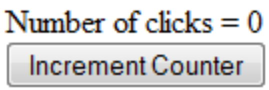
```
function handleAllTags() {
  const fontElements = document.querySelectorAll("font");
  alert(`Number of font tags in this document are ${fontElements.length}`);
}
```

JavaScript w/ jQuery

```
$(function () {
  // when document is ready
  $("#countTags").click(function () {
    // when countTags is clicked,
    alert("Number of font tags in this document are " + $("font").length);
    // alert the number of font tags in the HTML
  });
});
```

http://csci571.com/examples/dom/ex2.html
http://csci571.com/examples/jquery/dom/ex2.html

Example 3: document.getElementById().innerHTML



JavaScript w/o jQuery

```
var hits = 0;
function updateMessage() {
  document.getElementById("counter").innerHTML =
                                               `Number of clicks = ${++hits}`;
}
```

JavaScript w/ jQuery

```
$(function () {
  var hits = 0;
  $("#updateMessage").click(function () {
    $("#counter").html("Number of clicks = " + ++hits);
  });
});
```

http://csci571.com/examples/dom/ex3.html

http://csci571.com/examples/jquery/dom/ex3.html

Example 4: document.getElementById().style.left

JavaScript and HTML w/o jQuery

```
<form>
  <input id="counter1" style="position:relative; left:0px" type="button" value="Move Button
           right once" onclick="document.getElementById('counter1').style.left = '500px';">
</form>
<br><br><br><br><br><br><br><br><br>
<form>
  <input id="counter2" style="position:relative; top:0px" type="button" value="Move Button
              down Once" onclick="document.getElementById('counter2').style.top = '15px';">
</form>
```

http://csci571.com/examples/dom/ex4.html

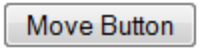
Example 4: $.css();

JavaScript and HTML w/ jQuery

```
<form>
  <input id="counter1" style="position:relative; left:0px" type="button" value="Move Button
       right once" onclick="$('#counter1').css('left', '500px');">
</form>
<br><br><br><br><br><br><br><br><br>
<form>
  <input id="counter2" style="position:relative; top:0px" type="button" value="Move Button
       down Once" onclick="$('#counter2').css('top', '15px');">
</form>
```

http://csci571.com/examples/jquery/dom/ex4.html

Example 5: document.getElementById(), parseint()



JavaScript w/o jQuery

```
var obj = document.getElementById("counter1");
var xlocation = parseInt(obj.style.left);
function handleClick() {
  xlocation += 50;
  document.getElementById("counter1").style.left = xlocation + "px";
}
```

JavaScript w/ jQuery

```
$(function () {
  $("#counter1").click(function () {
    const xlocation = parseInt($("#counter1").css("left")) + 50;
    $("#counter1").css("left", xlocation + "px");
  });
});
```

http://csci571.com/examples/dom/ex5.html
http://csci571.com/examples/jquery/dom/ex5.html

Example 6: Uses childNodes, removechild, appendChild

paragraph #1

paragraph #2

paragraph #3

Click Me to Reverse

JavaScript w/o jQuery

```
function reverse(n) {
  var kids = n.childNodes; // Get the list of children
  for (let i = kids.length - 1; i >= 0; i--) {
    // Loop backward through the children
    n.appendChild(n.removeChild(kids[i])); // Remove and reinsert
  }
}
```

http://csci571.com/examples/dom/ex6.html

Example 6: $.children(), $.remove(), $.append();

paragraph #1

paragraph #2

paragraph #3

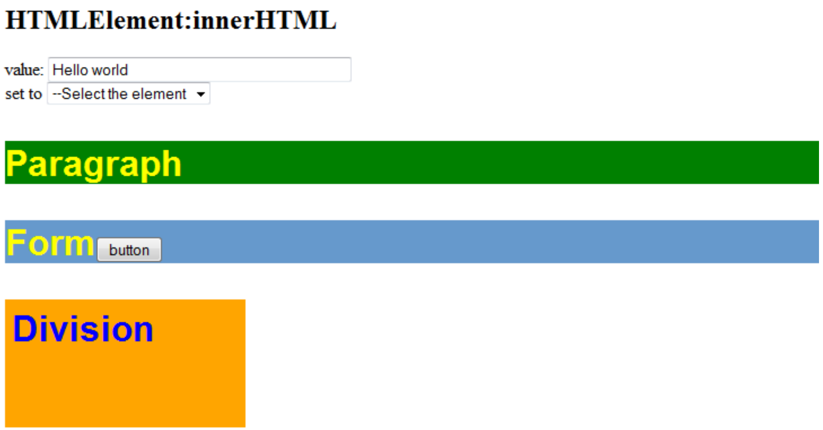
Click Me to Reverse

JavaScript w/ jQuery

```
var onReady = function () {
  $(".reverse").on("click", function () {
    var kids = $("body").children();
    for (var i = kids.length - 1; i >= 0; i--) {
      var c = $(kids[i]).remove();
      $("body").append(c);
    }
    onReady();
  });
};

$(onReady);
```

http://csci571.com/examples/jquery/dom/ex6.html

Example 4b: Uses innerHTML

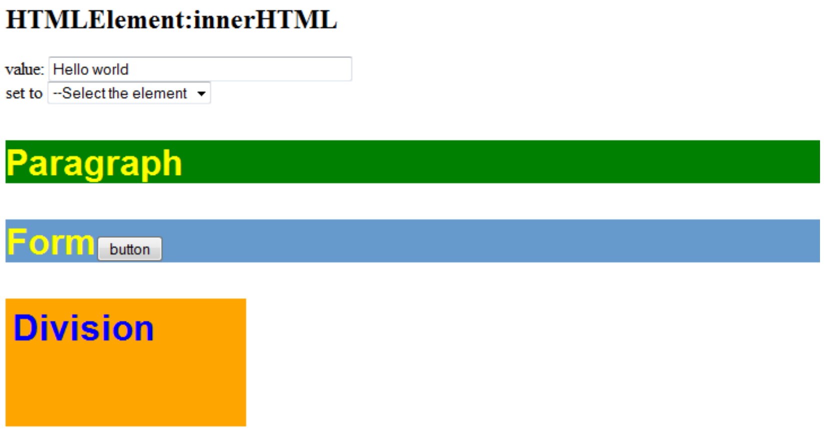


JavaScript w/o jQuery

```
function setInnerHTML(nm, value) {
  const element = document.querySelector(nm);
  if (!element) return;

  element.innerHTML = value;
}
```

http://csci571.com/examples/dom/domtest.html

Example 4b: $.change() and $.html();



JavaScript w/ jQuery

```
$(function () {
  $("#sel").change(function () {
    var selector = "#" + $("#sel").val();
    $(selector).html( $("input[name='t']").val() );
  });
});
```

http://csci571.com/examples/jquery/dom/ex4b.html

jQuery & AJAX

- jQuery has a series of functions which provide a common interface for AJAX, no matter what browser you are using.
- Most of the upper-level AJAX functions have a common layout:
  - **$.func**(url[,params][,callback]), [ ] optional
    - url: string representing server target
    - params: names and values to send to server
    - callback: function executed on successful communication.

jQuery AJAX load method

- The **load()** method loads data from a server and puts the returned data into the selected element.

```
$(selector).load(URL, [data, callback]);
```

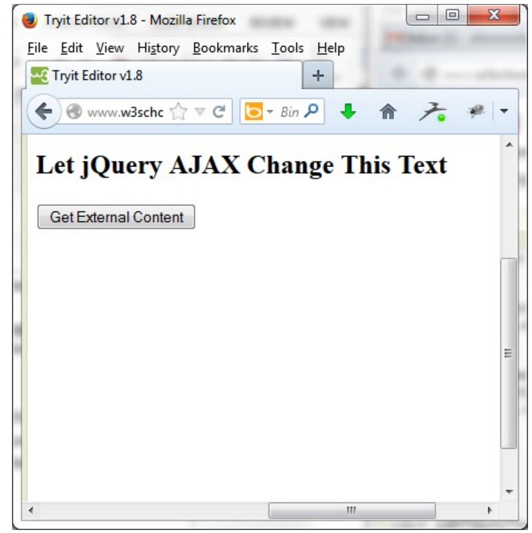
- The selector is a usually a reference to div or span tag
- The required *URL* parameter specifies the URL you wish to load.
- The optional *data* parameter specifies a set of querystring key/value pairs to send along with the request.
- The optional *callback* parameter is the name of a function to be executed after the load() method is completed.
- For examples see

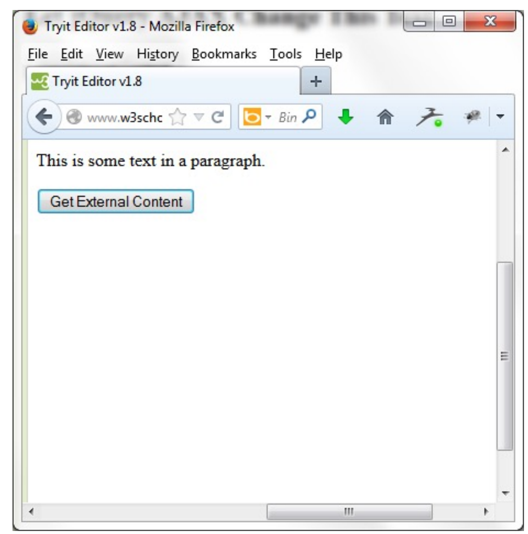
http://csci571.com/ajaxexamples/simple/simpleajaxexjquery.html

or http://csci571.com/examples.html#jquery [4 examples under Ajax]

AJAX Example 1

```
<!DOCTYPE html><html><head>
<script src="https://code.jquery.
         com/jquery-3.4.1.min.js">
</script>
<script>
$(document).ready(function(){
  $("button").click(function(){
    $("#div1")
       .load("demo_test.txt #p1");
  });
});
</script>
</head><body>
<div id="div1">
  <h2>Let jQuery AJAX Change This
                             Text</h2>
</div>
  <button>Get External
          Content</button>
</body></html>
```



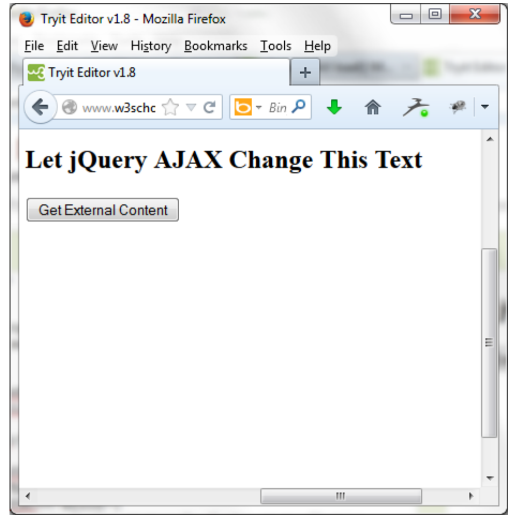


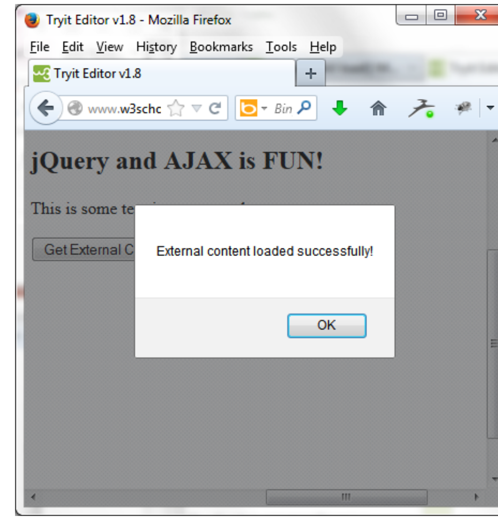
AJAX Example 2

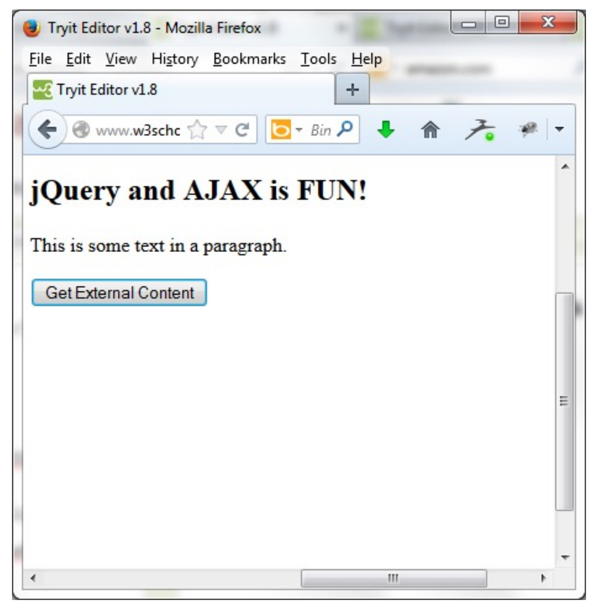
```
<!DOCTYPE html><html><head>
<script
src="https://code.jquery.com/jquery-
3.4.1.min.js">
</script><script>
$(document).ready(function(){
  $("button").click(function(){

$("#div1").load("demo_test.txt",function
(responseTxt,statusTxt,xhr){
      if(statusTxt=="success")
        alert("External content loaded
successfully!");
      if(statusTxt=="error")
        alert("Error: "+xhr.status+":
"+xhr.statusText);   });    });    });
</script></head><body>
<div id="div1"><h2>Let jQuery AJAX
Change This Text</h2></div>
<button>Get External
Content</button></body></html>
```

Note: non-jQuery version of .ready:

```
document.addEventListener("DOMContentLoaded",
function(event) { //do work });
```

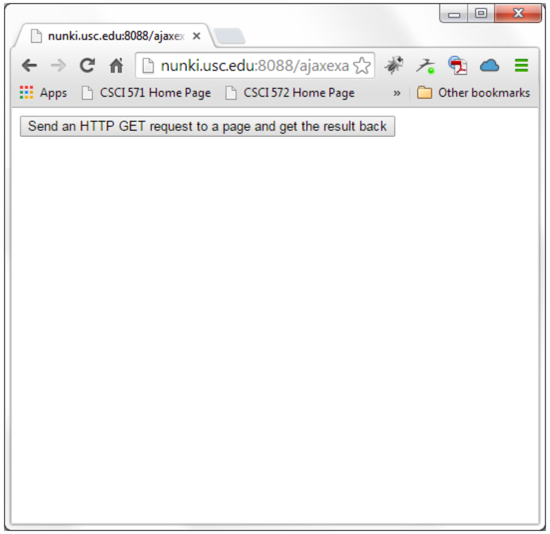


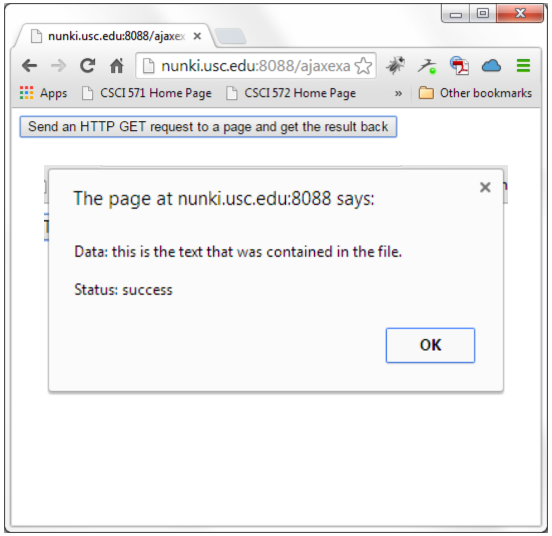




AJAX Example 3 – GET Method

```
<!DOCTYPE html><html><head>
<script
src="https://code.jquery.com/jquery-
3.4.1.min.js">
</script><script>
$(document).ready(function(){
  $("button").click(function(){

$.get("demo_test_get.pl",function(data,
status){
      alert("Data: " + data +
"\nStatus: " + status);
    });
  });
});
</script></head><body>
<button>Send an HTTP GET request to a
page and get the result back</button>
</body></html>
```



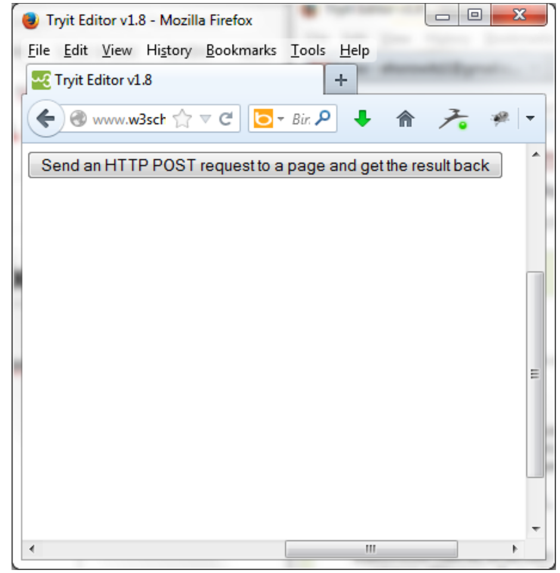


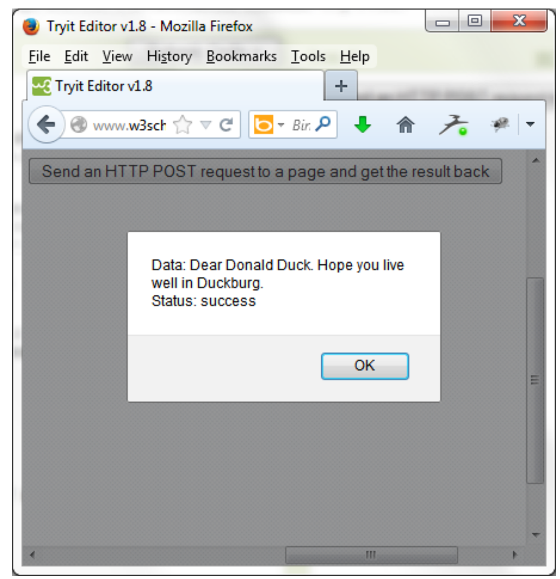
The $.get() method requests data from the server with an HTTP GET request.
The required URL parameter specifies the URL you wish to request.
The optional callback parameter is the name of a function to be executed if the request succeeds.
The following example uses the $.get() method to retrieve data from a file on the server

AJAX Example 4 – POST Method

```
<!DOCTYPE html><html><head>
<script src="https://code.jquery.com/jquery-3.4.1.min.js">
</script><script>
$(document).ready(function(){
  $("button").click(function(){
    $.post("demo_test_post.php",
    {
      name:"Donald Duck",
      city:"Duckburg"
    },
    function(data,status){
      alert("Data: " + data + "\nStatus: " + status);
    });
  });
});
</script></head><body>
<button>Send an HTTP POST request to a page and get the result back</button>
</body></html>
```

```
<?php
echo "Dear ".$_POST["name"]." Hope you live well in ".$_POST["city"];
?>
```





Summary jQuery AJAX Functions

- $.**func**(url[,params][,callback])
  - $.get
  - $.getJSON
  - $.getIfModified
  - $.getScript
  - $.post
- $(selector), inject HTML
  - load
  - loadIfModified
- $(selector), ajaxSetup alts
  - ajaxComplete
  - ajaxError
  - ajaxSend
  - ajaxStart
  - ajaxStop
  - ajaxSuccess
- **$.ajax**, $.**ajaxSetup**
  - async
  - beforeSend
  - complete
  - contentType
  - data
  - dataType
  - error
  - global
  - ifModified
  - processData
  - success
  - timeout
  - type
  - url

jQuery Usage Example (1) - Event

- jQuery way of a mouseover event that shows a submenu when menu is selected:

```
$("#menu").mouseover(function () {
    // Anonymous function
    $("#submenu").show();
});
```

jQuery Usage Example (2) - Event

- Stopping a normal **event** action: Suppose we want to stop the action of following a URL when a link is clicked. The action is part of the event object. We can reference the event object and call .preventDefault();

```
$('#menu').click(function(evt){
    //JS code here
    evt.preventDefault();
})
```

jQuery Usage Example (3) – Form Selectors

- Selecting all form elements of a certain type:
  `$(':text')` It selects all text fields.

- Use with `:input ( all form elements), :password, :radio, :checkbox, :submit, :image, :reset, :button, :file, :hidden`

  See https://api.jquery.com/category/selectors/form-selectors/

- Set the value of a form element
  ```
  let fieldvalue = $('#total').val(Yourvalue);
  ```
  See https://api.jquery.com/val/

jQuery Usage Example (4) - Attribute

- Determining if checkbox is checked

```
if ($("#total").attr("checked")) {
  //Do whatever you want if box is checked
} else {
  //Do whatever you want if box is not checked
}
```

jQuery Usage Example (5) – Form Events

- Form Events such as submit:

```
$(document).ready(function () {
   $("#signup").submit(function () {
      if ($("#username").val() == "") {
         alert("Please supply name to name field");
         return false;
      }
   });
});
```

jQuery Usage Example (6) - More events

- Focus Example: Auto erases default text in a field when it gets the focus

```
<input name="username" type="text" id="username"
       value="Please type your user name">

$('#username').focus(function () {
    var field = $(this);
    if (field.val() == field.attr('defaultValue')) {
        field.val('');
    }
});
```

jQuery Usage Example (7)

- Click: If any radio button is clicked

```
$(':radio').click(function () {
  //do stuff
});
```

- Add focus to the first element of the form:

```
$('username').focus();
```

Is jQuery Worth It?

Yes	No
Good use of the jQuery library will make it worthwhile in your code; will make JavaScript more readable and understandable	Bad use of jQuery library adds extra overhead. Why even add jQuery? Remember you need to add: `<script src="//ajax.googleapis.com/ajax/libs/jquery/1.8.3/jquery.min.js"></script>` Sometimes it is better to stick with WebStandards as they would outlive any library
If web application requires some imperative DOM manipulation, hiding elements, fading out elements, etc	Doesn’t even need DOM manipulation; could be done with CSS. If the application is large it is better to use declarative reactive frameworks!
Old Browser Support – no need extra code for browser compatibility	Modern browsers have almost perfect compatibility

You might not need JQuery!

- https://youmightnotneedjquery.com/

JQUERY

```
$(el).fadeIn();
```

MODERN

```
el.classList.replace('hide', 'show');
```

```
.show {
  transition: opacity 400ms;
}
.hide {
  opacity: 0;
}
```

JQUERY

```
$(el).addClass(className);
```

IE10+

```
el.classList.add(className);
```

JQUERY

```
$(el).attr('tabindex', 3);
```

IE8+

```
el.setAttribute('tabindex', 3);
```

jQuery

- It’s a useful library **when used wisely.**
- It will allow you to write JavaScript differently
  - **Write less, do more.**
- Remember: jQuery is just JavaScript
  - What you can do with jQuery, **you can always do without jQuery** but with ***more code.***

jQuery Resources

- Project website
  - http://www.jquery.com
- Learning Center
  - http://docs.jquery.com/Tutorials
  - http://www.learningjquery.com/
  - https://www.w3schools.com/jquery/default.asp
- Support
  - http://docs.jquery.com/Discussion
  - http://www.nabble.com/JQuery-f15494.html mailing list archive
  - irc.freenode.net irc room: #jquery
- Documentation
  - http://docs.jquery.com/Main_Page
  - http://www.visualjquery.com
  - http://jquery.bassistance.de/api-browser/
- jQuery Success Stories
  - http://docs.jquery.com/Sites_Using_jQuery
- jQuery selectors Demo
  - https://api.jquery.com/category/selectors/